5.5 Classwork: Proof, Differential Equations and Series — Markscheme

Total 59 marks.

1. [Maximum mark: 13]

(a) $n^3 - n = (n-1)n(n+1)$ **A1**

$n-1$ and n are consecutive integers, so one of them is even, and the product is even **R1**

Accept a case argument: $n = 2m$ gives $2m(4m^2-1)$; $n = 2m+1$ gives $(2m+1)(2m)(2m+2)$. Checking a few values of n is not a proof: A0R0.

(b) Assume there exists $\alpha \in \mathbb{Z}$ such that $\alpha^3 + \alpha + 1 = 0$ **M1**

M0 for vague openings such as "consider the equation has integer roots" or "let $\alpha \in \mathbb{Z}$ be a root"; the assumption must state that an integer root exists. Condone x in place of α. Later marks do not depend on this M1 (as in the IB markscheme for 2022 May P1 TZ2 Q9).

$\alpha^3 + \alpha = -1$ **M1**

$\alpha^3 + \alpha = (\alpha^3 - \alpha) + 2\alpha$ **A1**

$\alpha^3 - \alpha$ is even by part (a) and 2α is even, so $\alpha^3 + \alpha$ is even **R1**

but -1 is odd, a contradiction, so the equation has no integer roots **R1 AG**

Accept $\alpha^3 + \alpha = \alpha(\alpha^2+1)$ with the case argument (α even, or α odd so α^2+1 even) for A1R1. Accept the method of the increasing function: $f'(x) = 3x^2 + 1 > 0$, so f is increasing; $f(-1) = -1$ and $f(0) = 1$, so the only real root lies between -1 and 0, which contradicts the assumption: M1 (assumption) M1A1 (derivative and increasing) R1 (the two values) R1 (contradiction).

(c) Assume there exist $p, q \in \mathbb{Z}$ such that $p^2 - 8q - 3 = 0$ **M1**

$p^2 = 8q + 3 = 2(4q+1) + 1$, which is odd **A1**

so p is odd (if p were even, p^2 would be even) **R1**

let $p = 2k+1$, $k \in \mathbb{Z}$: $4k^2 + 4k + 1 = 8q + 3$ **M1**

$4k^2 + 4k - 8q = 2$

$2(k^2 + k - 2q) = 1$ **A1**

the left side is even and 1 is odd, a contradiction, so $p^2 - 8q - 3 \neq 0$ **R1 AG**

Accept $4k(k+1) = 8q + 2$: by part (a) reasoning $k(k+1)$ is even, so the left side is a multiple of 8, but $8q+2$ is not. Accept $k^2 + k - 2q = \frac{1}{2}$, not an integer.

2. [Maximum mark: 17]

(a) $\int \left(x^2+1\right) e^x \,\mathrm{d}x = \left(x^2+1\right) e^x - \int 2xe^x \,\mathrm{d}x$ **M1A1**

$\int 2xe^x \,\mathrm{d}x = 2xe^x - \int 2e^x \,\mathrm{d}x = 2xe^x - 2e^x$ **M1**

$\left(x^2-2x+3\right) e^x + C$ **A1**

Accept an unsimplified equivalent. Condone a missing +C here; it is needed in (b).

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = x^2+1$; integrating factor $e^{\int 1\,\mathrm{d}x} = e^x$ **M1**

$\dfrac{\mathrm{d}}{\mathrm{d}x}\left(ye^x\right) = \left(x^2+1\right) e^x$ **A1**

$ye^x = \left(x^2-2x+3\right) e^x + C$ **A1**

$x=0,\ y=1$: $1 = 3 + C$, so $C = -2$ **M1**

$y = x^2 - 2x + 3 - 2e^{-x}$ **A1 AG**

The final A1 needs the division by e^x applied to the constant: Ce^{-x}. Common error: $y = x^2-2x+3+C$ (constant not divided by e^x) leads to $y = x^2-2x+1$, so the AG is not reached: A0 for the last two marks unless the error is corrected. FT the integral from (a).

(c) $y'(0) = 0 + 1 - 1 = 0$ **A1**

$y'' = 2x - y'$ **M1**

$y''(0) = 0$ **A1**

$y''' = 2 - y''$, so $y'''(0) = 2$ **A1**

$y = 1 + \dfrac{2}{3!}x^3 = 1 + \dfrac{x^3}{3}$ **A1**

Accept $1 + 0x + 0x^2 + \frac{1}{3}x^3$. Expanding the solution from (b) instead is not what the question asks: M0, but award the final A1 for $1 + \frac{x^3}{3}$.

(d) $y' = 2x - 2 + 2e^{-x},\ \ y'' = 2 - 2e^{-x}$ **M1**

$y''(0) = 2 - 2 = 0$ **A1**

$y'' < 0$ for $x < 0$ (since $e^{-x} > 1$) and $y'' > 0$ for $x > 0$, so the concavity changes at $x = 0$; and $y(0) = 3 - 2 = 1$ **R1**

R1 needs the sign change of y'', not only $y''(0) = 0$. Accept the series from (c): near 0, $y \approx 1 + \frac{x^3}{3}$, so $y'' \approx 2x$, which changes sign at 0.

3. [Maximum mark: 15]

(a) $n = 1$: $f'(x) = \dfrac{1}{1+x}$ and $\dfrac{(-1)^0\,0!}{(1+x)^1} = \dfrac{1}{1+x}$, so true for $n = 1$ **A1**

assume true for $n = k$: $f^{(k)}(x) = \dfrac{(-1)^{k-1}(k-1)!}{(1+x)^k}$ **M1**

Do not accept "let $n = k$" or "assume $n = k$"; the statement must be assumed true.

$f^{(k+1)}(x) = \dfrac{\mathrm{d}}{\mathrm{d}x}\left[(-1)^{k-1}(k-1)!\,(1+x)^{-k}\right]$ **M1**

$= (-1)^{k-1}(k-1)!\,(-k)(1+x)^{-k-1}$ **A1**

$= \dfrac{(-1)^k\,k!}{(1+x)^{k+1}}$ **A1**

this is the statement for $n = k+1$ **R1**

since true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, the statement is true for all $n \in \mathbb{Z}^+$ **R1**

The final R1 needs both the base case and the inductive step to have been attempted.

(b) $f(0) = 0$ and $f^{(n)}(0) = (-1)^{n-1}(n-1)!$ **M1**

coefficient of x^n: $\dfrac{(-1)^{n-1}(n-1)!}{n!}$ **A1**

$= \dfrac{(-1)^{n-1}}{n}$, so $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^{n-1}x^n}{n} + \cdots$ **A1 AG**

(c) $\ln\left(\dfrac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$ **M1**

$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots$ **A1**

$2x + \dfrac{2x^3}{3} + \dfrac{2x^5}{5}$ **A1**

Common error: $\ln\left(\frac{1+x}{1-x}\right)$ *written as* $\frac{\ln(1+x)}{\ln(1-x)}$, *or* $\ln(1+x)$ *split as* $\ln 1 + \ln x$*: M0.*

(d) $\dfrac{\frac{2x^3}{3} + \frac{2x^5}{5} + \cdots}{x^3} = \dfrac{2}{3} + \dfrac{2x^2}{5} + \cdots$ **M1**

limit $= \dfrac{2}{3}$ **A1**

FT from (c). Accept l'Hôpital's rule applied correctly.

4. [Maximum mark: 14]

(a) $w + w^* = (a + bi) + (a - bi) = 2a = 2\,\mathrm{Re}(w)$ **A1**

$ww^* = (a + bi)(a - bi) = a^2 - b^2 i^2$ **M1**

$= a^2 + b^2 = |w|^2$ **A1 AG**

(b) $w^* = re^{-i\theta}$ **A1**

$(w^*)^n = r^n e^{-in\theta}$ **M1**

$w^n = r^n e^{in\theta}$, so $(w^n)^* = r^n e^{-in\theta} = (w^*)^n$ **A1 AG**

Accept the same argument in cis form: $(r\,\mathrm{cis}(-\theta))^n = r^n\,\mathrm{cis}(-n\theta)$.

(c) $1 - i = (1 + i)^*$, so $(1 - i)^n = ((1 + i)^n)^*$ by (b) **R1**

$(1 + i)^n + (1 - i)^n = z + z^* = 2\,\mathrm{Re}(z)$, where $z = (1 + i)^n$, which is real by (a) **R1 AG**

(d) $1 + i = \sqrt{2}\,e^{i\frac{\pi}{4}}$ **A1**

$(1 + i)^n = 2^{\frac{n}{2}} e^{i\frac{n\pi}{4}}$ **M1**

$(1 + i)^n + (1 - i)^n = 2\,\mathrm{Re}\left(2^{\frac{n}{2}} e^{i\frac{n\pi}{4}}\right)$ **M1**

$= 2 \times 2^{\frac{n}{2}} \cos\dfrac{n\pi}{4} = 2^{\frac{n}{2}+1} \cos\dfrac{n\pi}{4}$ **A1 AG**

Accept adding $2^{\frac{n}{2}} e^{i\frac{n\pi}{4}}$ *and* $2^{\frac{n}{2}} e^{-i\frac{n\pi}{4}}$ *directly, the sine terms cancelling, for the second M1.*

(e) $2^7 \cos 3\pi$ **M1**

-128 **A1**

Accept $(1+i)^2 = 2i$*, so* $(1+i)^{12} = (2i)^6 = -64$*, and the same for* $(1-i)^{12}$*, giving* -128.